Roll No. | | | | | | | | | | |

Total No. of Pages : 03

Total No. of Questions : 13

B.Pharmacy (Sem.-8)

BIOSTATISTICS AND RESEARCH METHODOLOGY

Subject Code : BP-801T

M.Code : 79764

Date of Examination : 24-11-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION - A

1. **Answer briefly :**

   a) Define Median.

   b) Calculate Standard deviation for the given data : 6, 8, 10, 7 and 9.

   c) Define Null hypothesis.

   d) Write any two importance's of multiple regression.

   e) Define Cohort studies.

   f) Write any two advantages of SPSS.

   g) Define degrees of freedom.

   h) Write any two assumptions for Poisson's distribution.

   i) Define Confidence interval.

   j) Define Range with example.

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SECTION - B

2. A tea company appoints four salesmen A, B, C and D and observes their sales in three seasons summer winter and monsoon. The data is given below

Seasons	**Salesmen**				**Season's total**
	A	**B**	**C**	**D**	
Summer	36	36	21	35	128
Winter	28	29	31	32	120
Monsoon	26	28	29	29	112
Salesmen's total	90	93	81	96	360

Perform the Two Way **ANOVA** for the above data :
(For (3, 6) d.f F0.05 = 4.76 ; For (2,6) d.f F0.05 = 5.14.)

3. Discuss in detail about designing heads of Clinical Trials With suitable examples

4. a) ***"A coin is tossed for six time".*** What is the probability of obtaining at least four heads?

   b) Describe briefly about multiple correlation.

SECTION - C

5. Write a note on various Graphs used in biostatistics to represent the data.

6. Describe the various types of sampling in Biostatistics.

7. **Perform the Kruskal wali test for the following given data :**

P :	5	7	8	10	11
Q :	10	16	12	13	12
R :	9	8	10	16	12

8. Give a note on factorial design and its advantages

9. Discuss in brief about Regression.

10. **Calculate Karl Pearson coefficient from the given data :**

S. No	1	2	3	4	5
Marks in Biostatistics	48	35	17	23	47
Marks in Product Development	45	20	40	25	45

11. Write a short note on any two software's used in Biostatistics and its importance.

12. **Calculate Mean and Mode for the following Data :**

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	4	3	9	10	2

13. Two types of drugs were used or 5 and 7 patients for reducing their weight. Drug A was imported and Drug B is from Dr. Reddys Pharma. **The Decrease in weight after using six months as follows :**

Drug A	10	12	13	11	14		
Drug B	8	9	12	14	15	10	9

Is there a significant difference in the efficacy of the two drugs? (For v=10 to.os is 2.23)

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.